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(19) **United States**(12) **Patent Application Publication**
SASAKI et al.(10) **Pub. No.: US 2025/0276834 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **DISCHARGE CAP****Publication Classification**(71) Applicant: **TENGA CO., LTD.**, Tokyo (JP)(72) Inventors: **Hajime SASAKI**, Tokyo (JP); **Shouji UEHIRA**, Tokyo (JP); **Satoshi ICHIKAWA**, Tokyo (JP)(51) **Int. Cl.****B65D 47/20** (2006.01)**B65D 47/08** (2006.01)**B65D 47/12** (2006.01)(52) **U.S. Cl.****CPC** **B65D 47/2031** (2013.01); **B65D 47/088** (2013.01); **B65D 47/128** (2013.01)(73) Assignee: **TENGA CO., LTD.**, Tokyo (JP)(21) Appl. No.: **18/879,692**(22) PCT Filed: **Apr. 6, 2023**(86) PCT No.: **PCT/JP2023/014259**

§ 371 (c)(1),

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ABSTRACT

The present invention has an object to suppress an external force applied to a container to discharge a content liquid CLq to an appropriate magnitude, to prevent the content liquid CLq discharged from a discharge nozzle 29 from becoming too thick, and to enable the content liquid CLq after discharge to be cut off in an improved manner. A valve part 49 included in a slit valve 7A includes a slit part 49b including slits 55, and an outer circumferential part 49a coupled to a thin-walled tube part 51 (a connecting part). The slit part 49b more easily deforms with a pressure change in the interior space of the container than the outer circumferential part 49a. The slit part 49b deforms in a discharge direction of the content liquid CLq at the time of discharge of the content liquid CLq, and deforms toward a nozzle flow path 29a after the content liquid CLq is discharged.

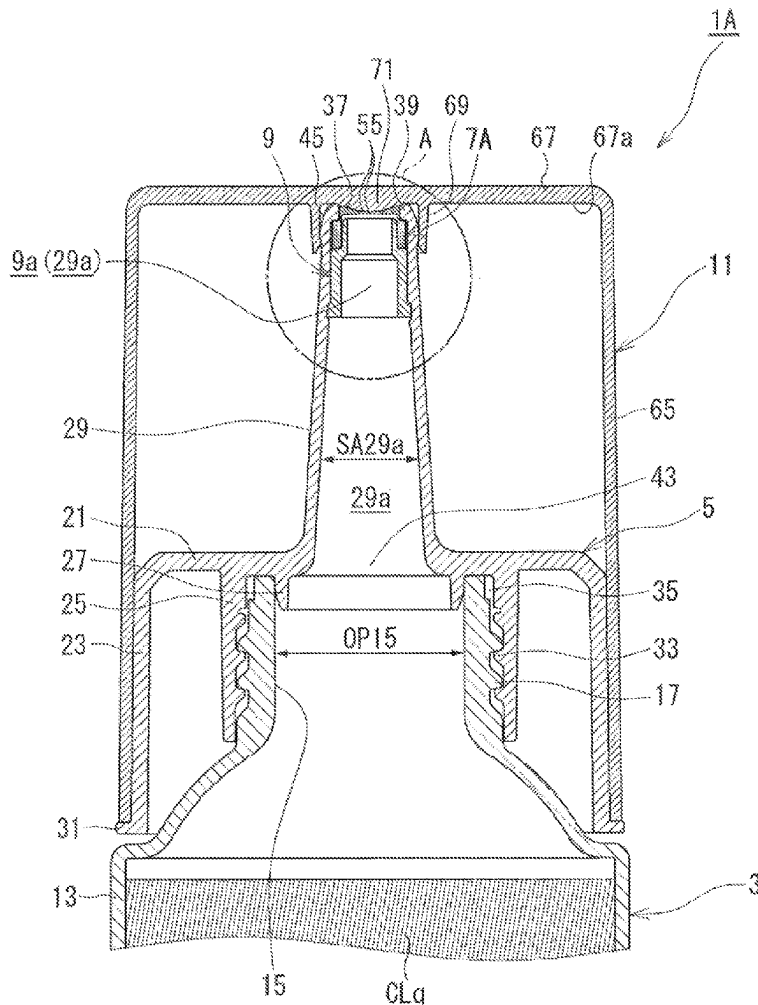


FIG.2

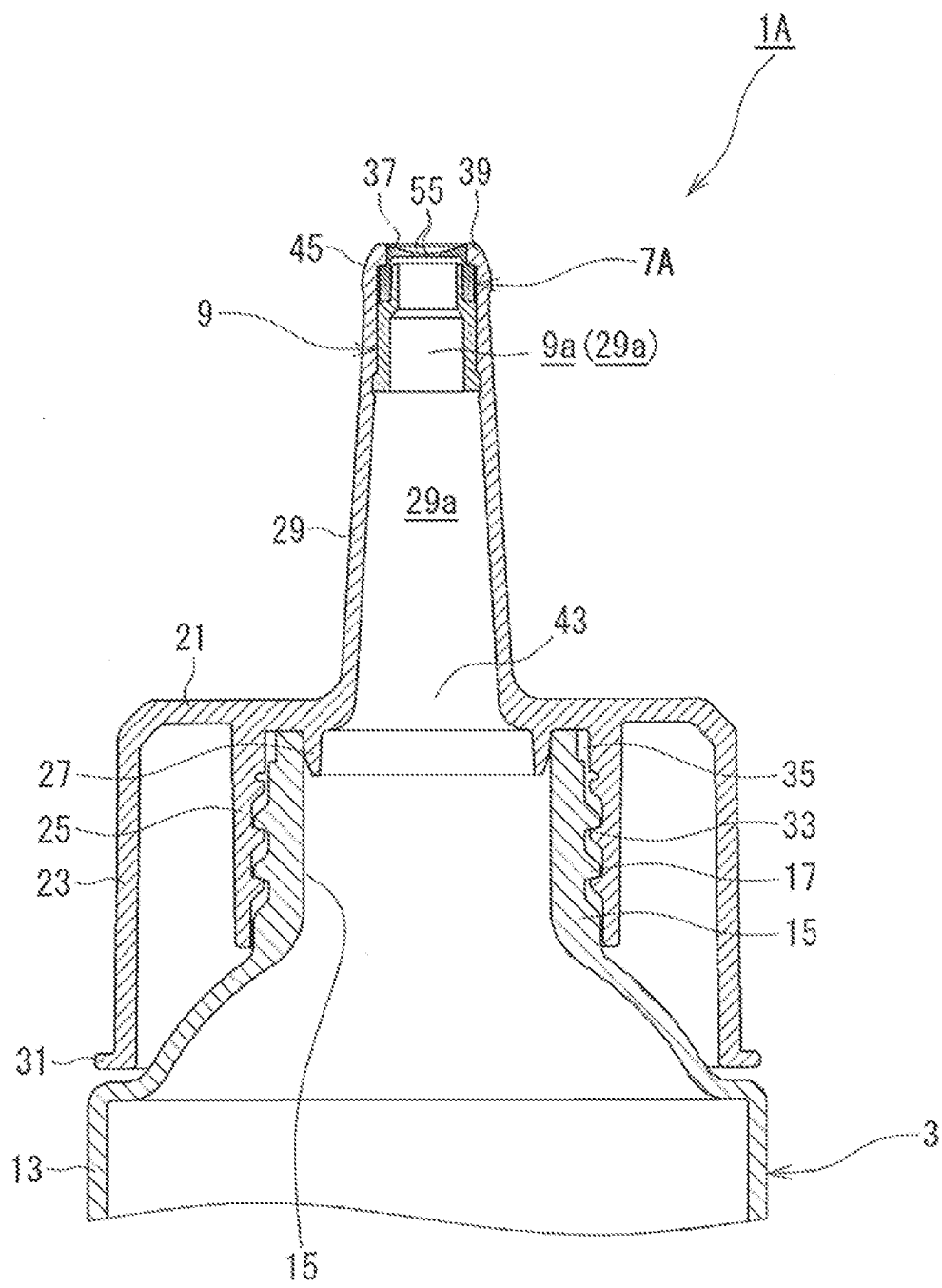


FIG.3

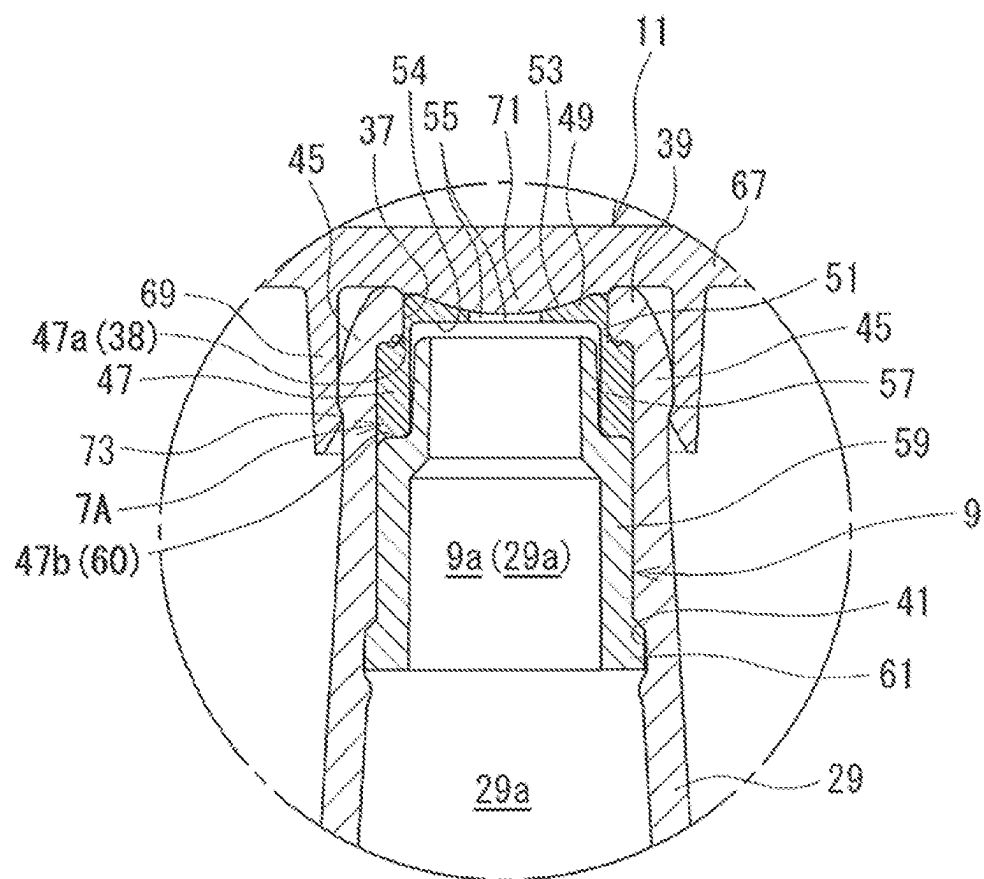


FIG.4A

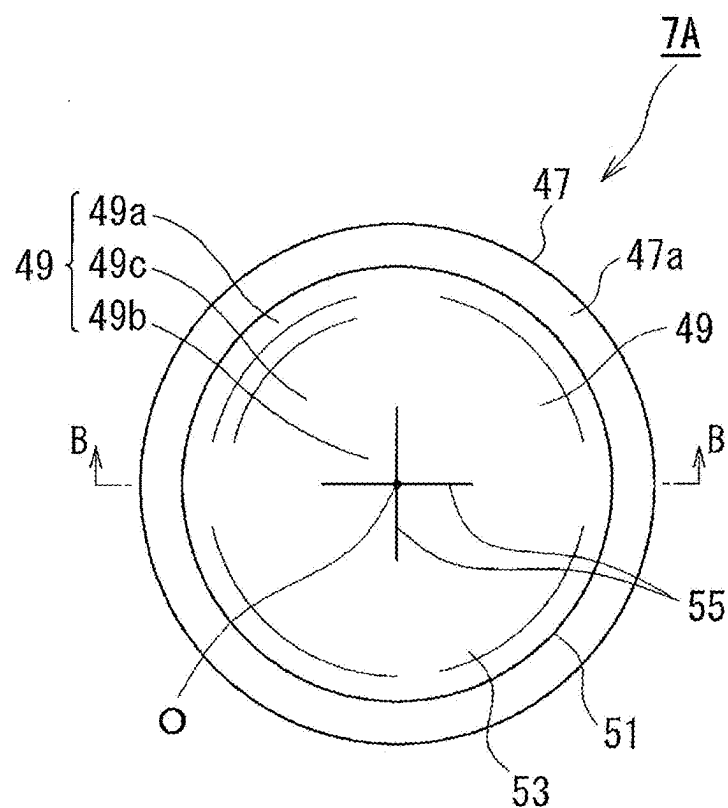


FIG.4B

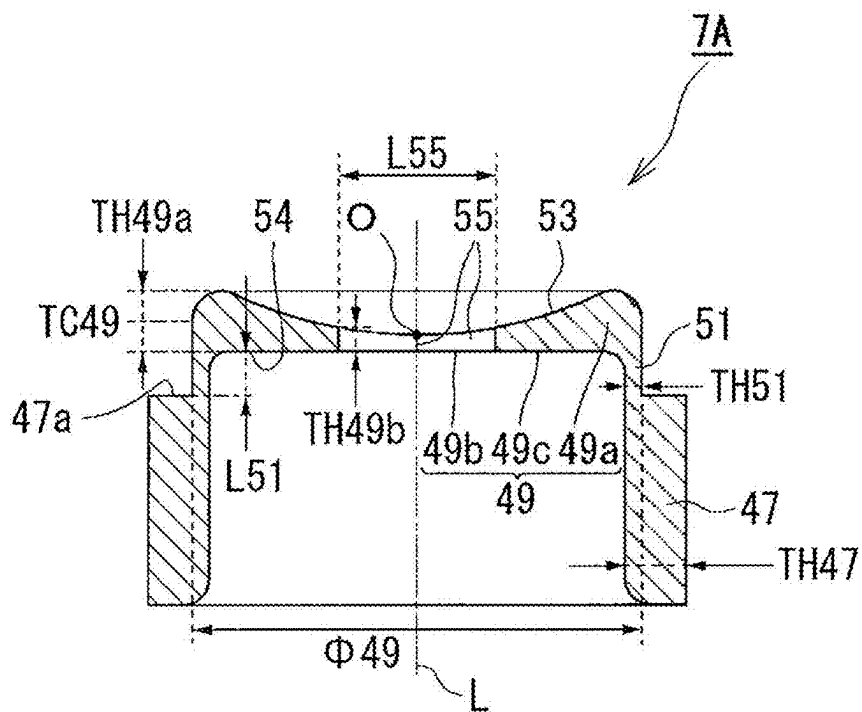


FIG.5A

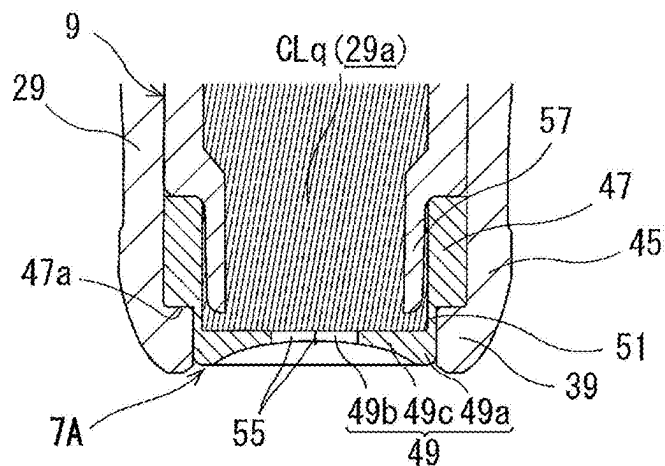


FIG.5B

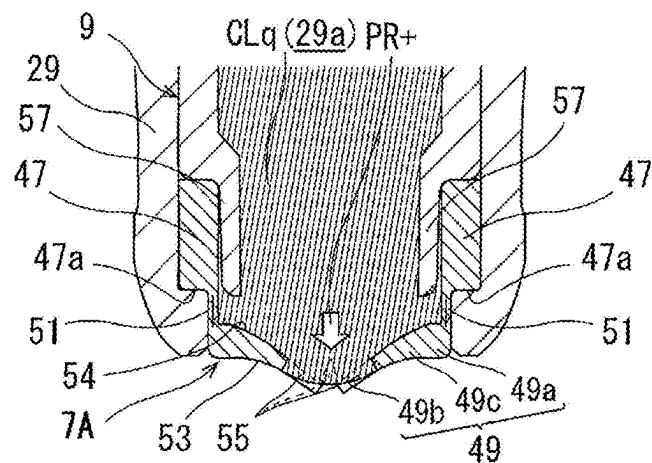


FIG.5C

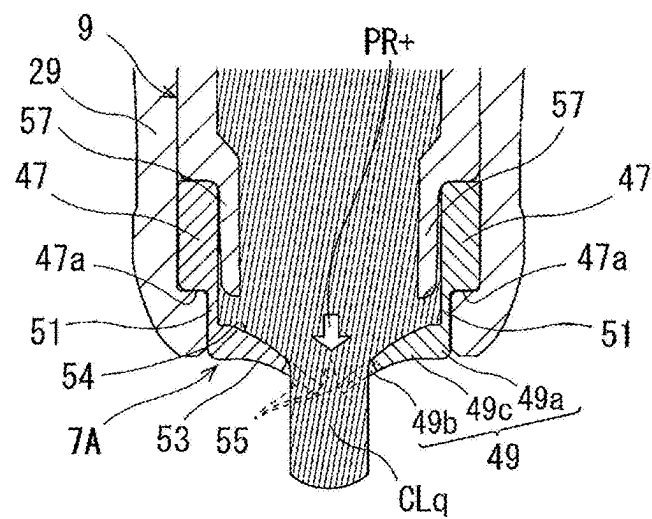


FIG. 6A

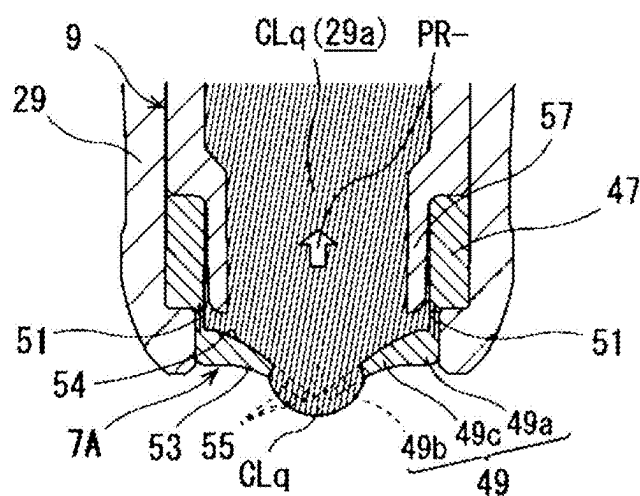


FIG. 6B

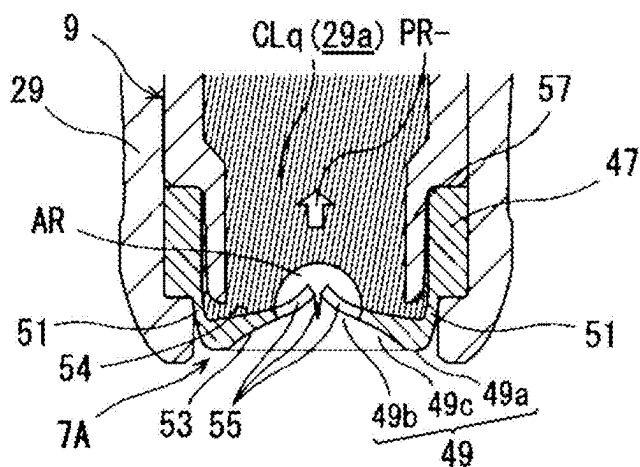


FIG. 6C

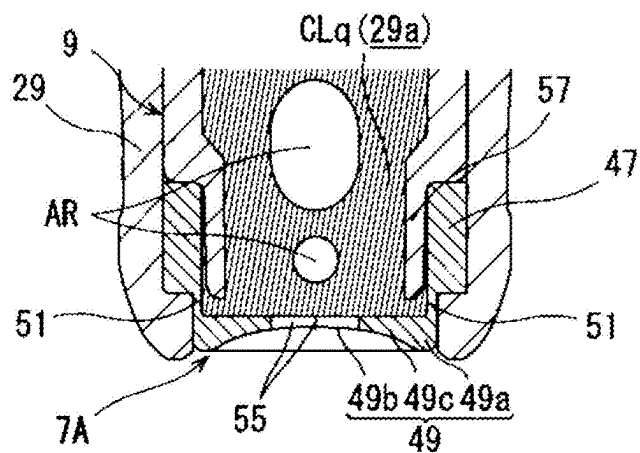


FIG.7A

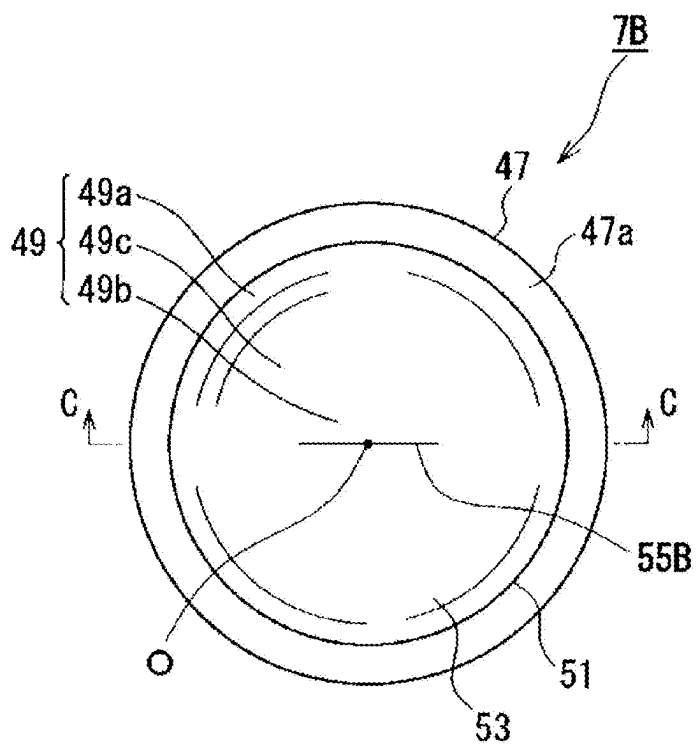


FIG.7B

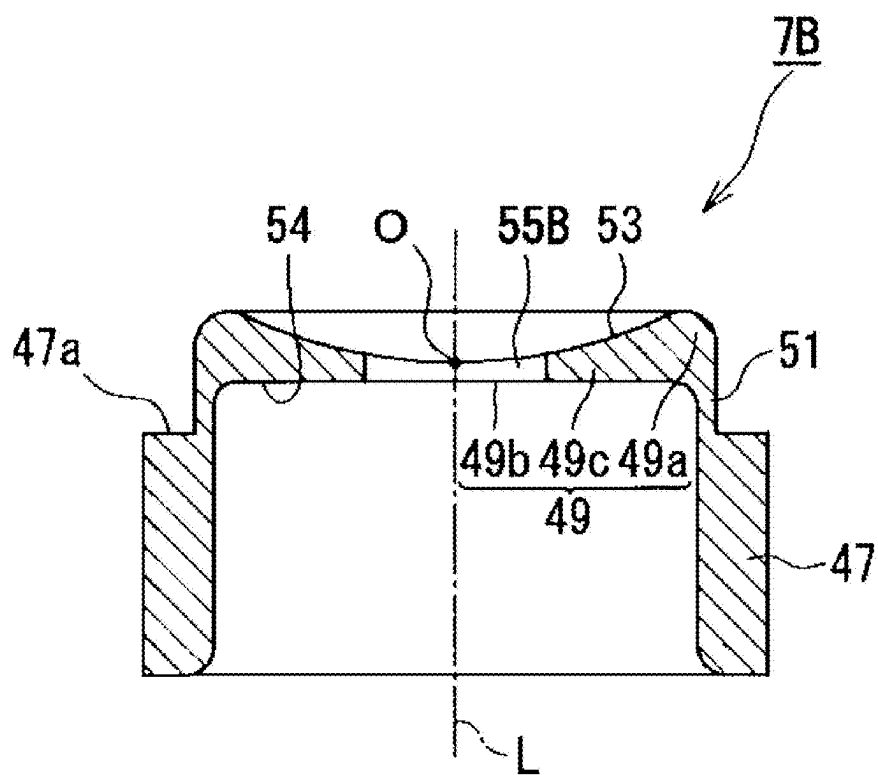


FIG.8

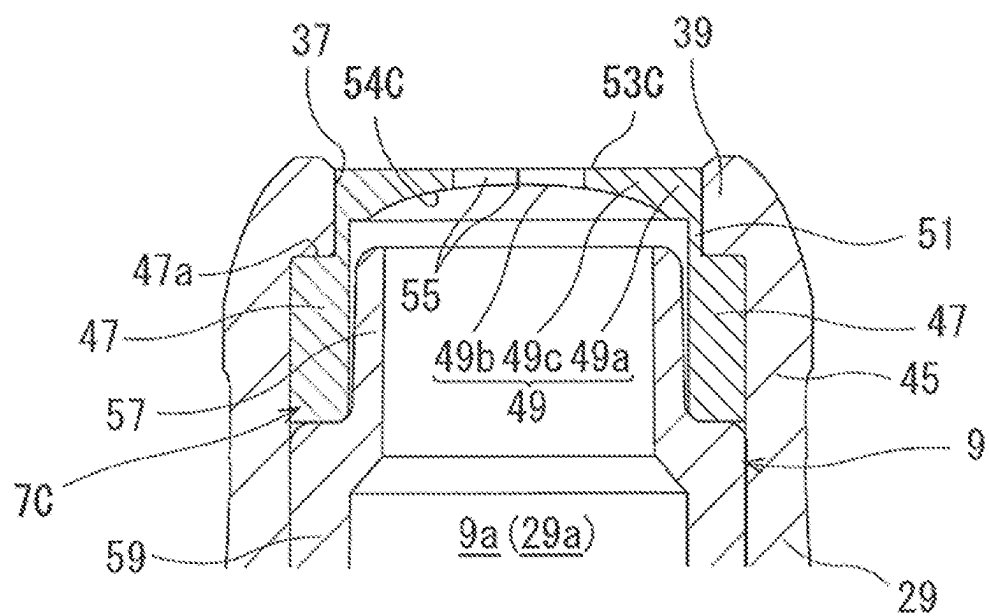


FIG.9

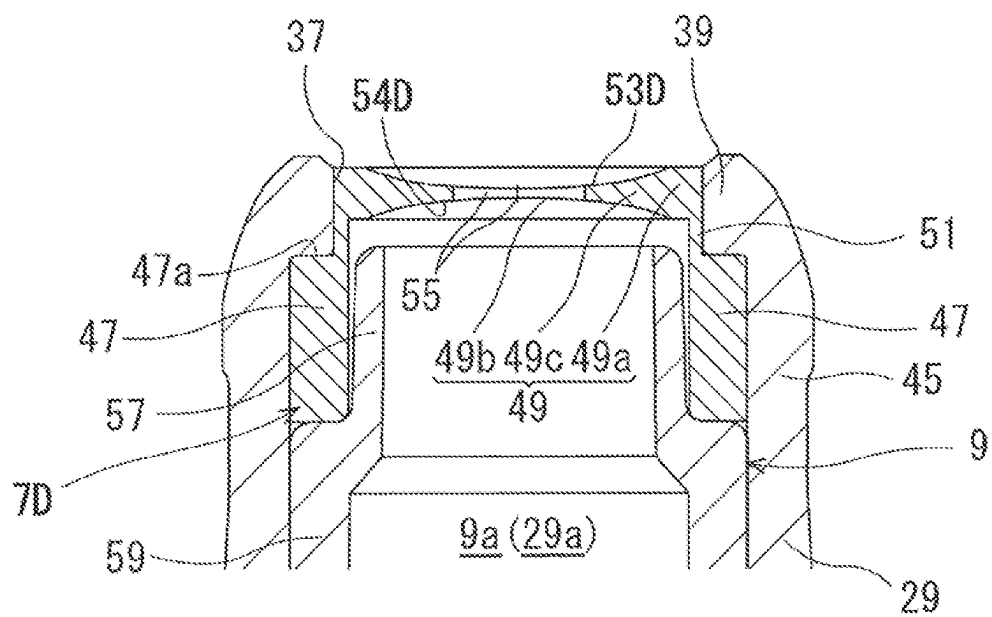


FIG.10

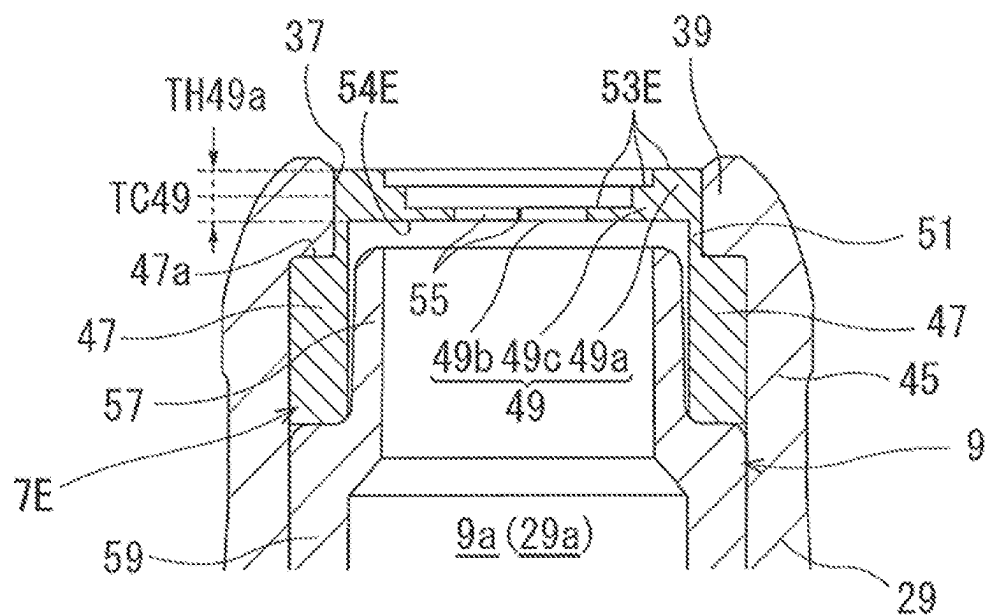


FIG.11

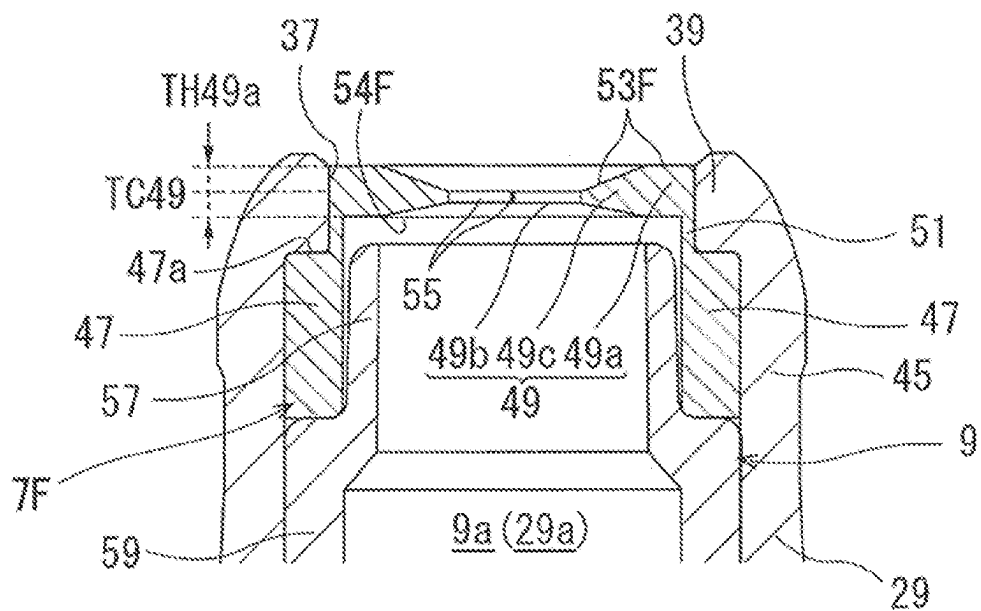


FIG.13A

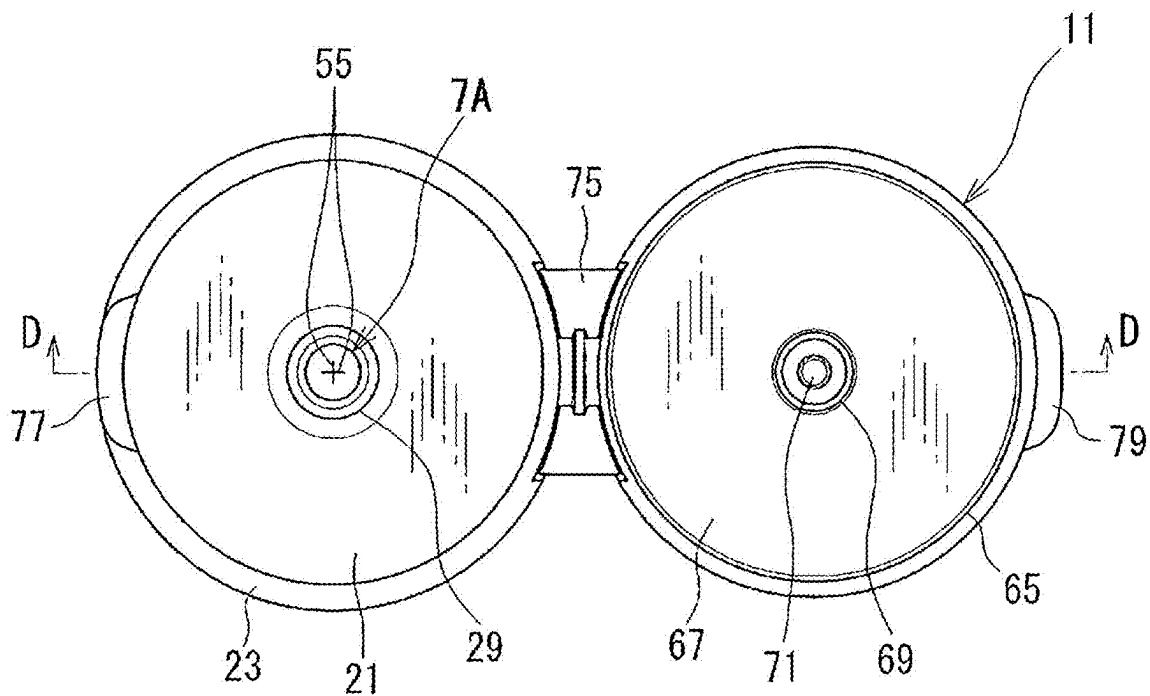


FIG.13B

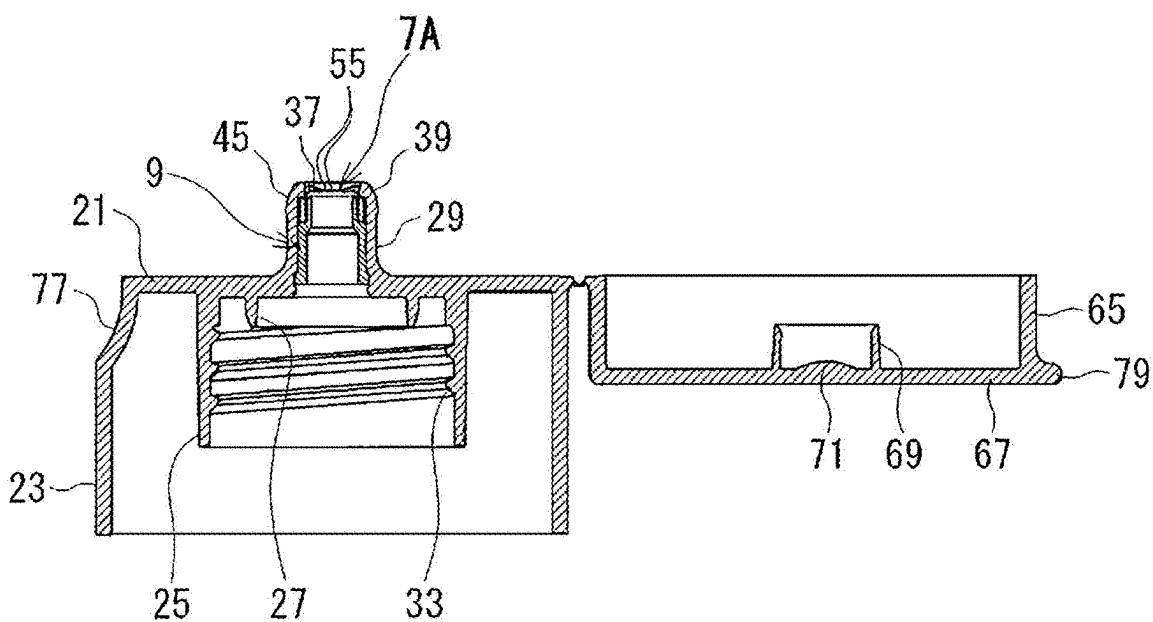


FIG.14

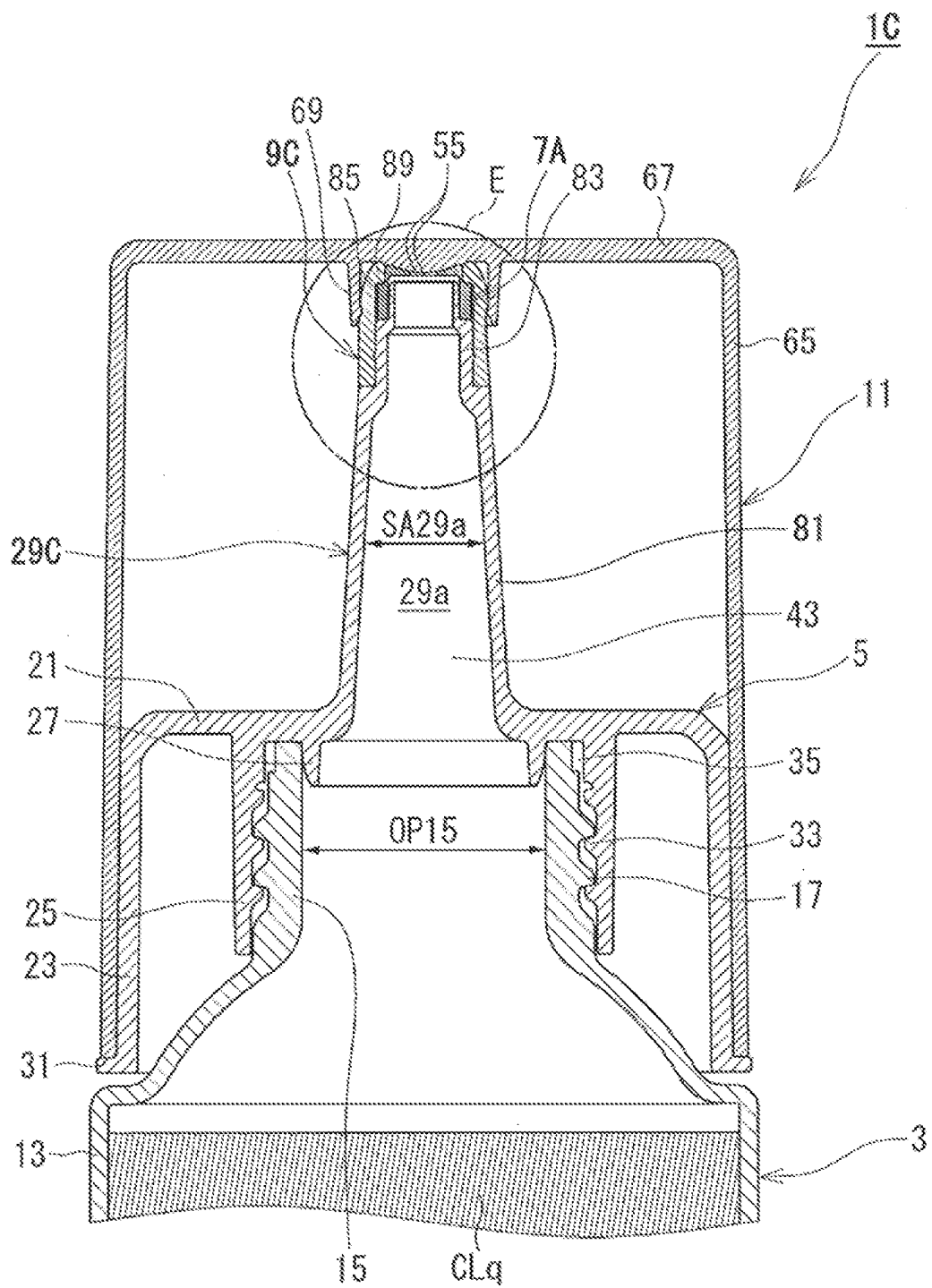


FIG. 15

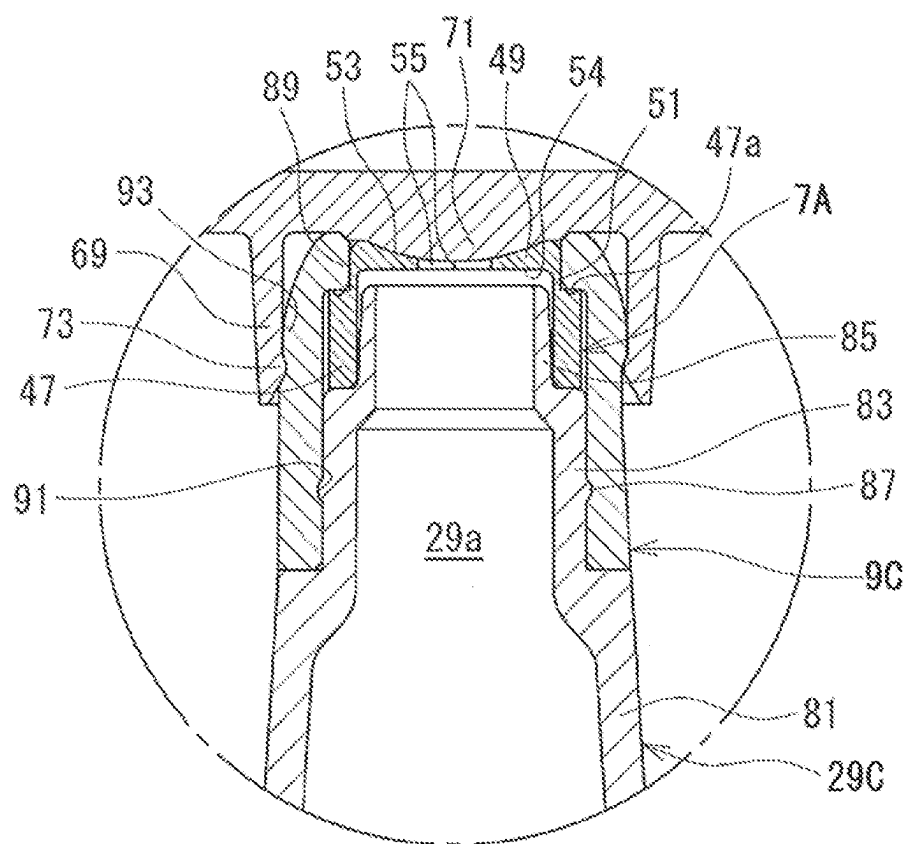
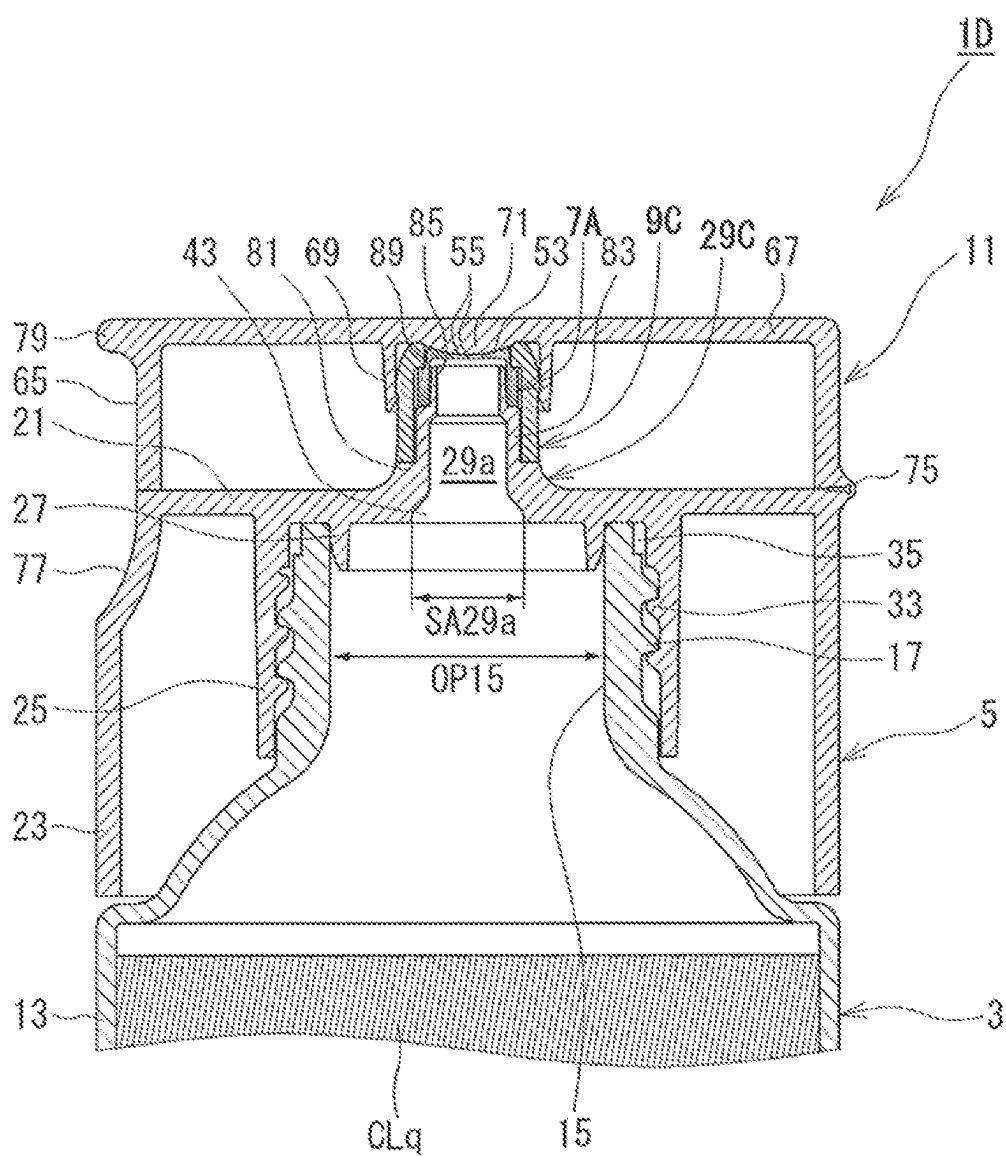


FIG. 16



DISCHARGE CAP**TECHNICAL FIELD**

[0001] The present invention relates to a discharge cap.

BACKGROUND

[0002] A viscous liquid, such as a lubricant, a caulking agent, or honey is stored in, for example, a container that has an elasticity and to which a discharge cap is attached at a mouth part. The discharge cap is provided with a nozzle for discharging a liquid (a content liquid) stored in the interior space of the container. The nozzle is a hollow tubular body where a flow path (a nozzle flow path) for a liquid is formed in the inner part thereof. The nozzle flow path is communicated with the interior space of the container when the discharge cap is attached to the mouth part of the container.

[0003] An external force is applied to the container when the content liquid is to be discharged from the nozzle. For example, when the container is compressed after the nozzle flow path is filled with the content liquid, the pressure in the interior space of the container is increased and the content liquid is discharged from the nozzle.

[0004] As such a discharge cap, for example, a discharge cap described in Patent Literature 1 is known. In the discharge cap described in Patent Literature 1, a nozzle includes a slit valve having slits in the shape of a cross formed thereon.

[0005] When the discharge cap in Patent Literature 1 is attached to a mouth part of a container, the nozzle flow path is filled with a content liquid, and an external force is then applied to the container, the content liquid in the nozzle flow path is pushed toward the slit valve with a pressure increase in the interior space of the container, and the slits in the shape of a cross are opened by the content liquid. Subsequently, the content liquid is discharged outside through a gap (a discharge port) of the slits.

[0006] When the external force to the container is canceled after the content liquid is discharged, the container inflates due to the elasticity of the container and the content liquid and air near the slits are drawn (sucked back) into the nozzle flow path. When the container returns to the original shape, the slits are closed.

[0007] In a container to which the discharge cap is attached, an external force to discharge the content liquid is required not to be too large. For example, even when the content liquid is a high-viscosity liquid such as a lubricant or honey, it is required that the content liquid is discharged with application of an appropriate magnitude of an external force to the container.

[0008] Elongating the slits provided in a valve to suppress the magnitude of an external force applied to the container is conceivable. However, simply elongating the slits may cause another problem that the content liquid discharged from the nozzle becomes too thick. Furthermore, a problem that the content liquid is not well cut off after discharge may also occur. For example, after the external force to the container is canceled, the content liquid that cannot be returned to the nozzle flow path may form a thread and adhere to the outer surface of the cap.

CITATION LIST**Patent Literature**

[0009] Patent Literature 1: Japanese Patent No. 6345040

SUMMARY**Technical Problem**

[0010] As described above, the conventional discharge caps have a problem that a content liquid discharged from a nozzle becomes too thick when slits are elongated to suppress an external force applied to a container.

[0011] The present invention has been made in view of the problems described above, and a primary object thereof is to suppress an external force applied to a container to discharge a content liquid to an appropriate magnitude, to prevent the content liquid discharged from a nozzle from becoming too thick, and to enable the content liquid after discharge to be cut off in an improved manner.

Solution to Problem

[0012] In order to solve the above problem, the present invention provides a discharge cap that is attached to a mouth part of a container having an elasticity and that discharges a content liquid associated with a pressure increase in an interior space of the container due to an external force applied to the container, the discharge cap comprising: a nozzle that forms a nozzle flow path with a sectional area smaller than an opening area of the mouth part; and a valve that is provided on the nozzle and that is made of an elastic material softer than the nozzle, wherein the valve includes a valve part through which the content liquid in the nozzle flow path is discharged, a tubular held part that is held on the nozzle by a holder, and a tubular connecting part that connects the valve part and the held part to each other, the valve part includes a slit part provided with a slit opening and closing according to a difference between an atmospheric pressure and a pressure in the interior space of the container, and an outer circumferential part coupled to the connecting part, and the slit part more easily deforms with a pressure change in the interior space than the outer circumferential part, deforms in a discharge direction of the content liquid at a time of discharge of the content liquid, and deforms toward the nozzle flow path after the content liquid is discharged.

Advantageous Effects of Invention

[0013] According to the present invention, the external force applied to the container to discharge the content liquid can be suppressed to an appropriate magnitude, the content liquid discharged from the nozzle is prevented from becoming too thick, and the content liquid after discharge is enabled to be cut off in an improved manner.

BRIEF DESCRIPTION OF DRAWINGS

[0014] FIG. 1 is a sectional view illustrating a discharge cap and a part of a container according to a first embodiment, and illustrating a closed state where a lid body is placed thereon.

[0015] FIG. 2 is a sectional view illustrating the discharge cap and a part of the container according to the first embodiment, and illustrating an open state where the lid body is removed therefrom.

[0016] FIG. 3 is an enlarged view of a portion A illustrated in FIG. 1.

[0017] FIG. 4 (a) is a plan view of a slit valve and FIG. 4 (b) is a sectional view along a line B-B illustrated in FIG. 4 (a).

[0018] FIG. 5 (a) is a view illustrating a state where a nozzle flow path is filled with a content liquid and no external force is applied to the container, FIG. 5 (b) is a view illustrating a state where a slit part of a slit valve is pushed and spread by the content liquid in the nozzle flow path, and FIG. 5 (c) is a view illustrating a state where the content liquid is discharged outside from the slit part.

[0019] FIG. 6 (a) is a view illustrating a state where the content liquid in the nozzle flow path starts returning to the container in a state where the external force to the container is canceled, FIG. 6 (b) is a view illustrating a state where air starts flowing into the nozzle flow path, and FIG. 6 (c) is a view illustrating a state where the flow of the air into the nozzle flow path ends and the slit valve is closed.

[0020] FIG. 7 (a) is a plan view of a slit valve according to a first variation, and FIG. 7 (b) is a sectional view along a line C-C illustrated in FIG. 7 (a).

[0021] FIG. 8 is a sectional view of a slit valve according to a second variation.

[0022] FIG. 9 is a sectional view of a slit valve according to a third variation.

[0023] FIG. 10 is a sectional view of a slit valve according to a fourth variation.

[0024] FIG. 11 is a sectional view of a slit valve according to a fifth variation.

[0025] FIG. 12 is a sectional view of a discharge cap according to a modification of the first embodiment.

[0026] FIG. 13 (a) is a plan view illustrating a discharge cap according to the modification of the first embodiment in an open state, and FIG. 13 (b) is a sectional view along a line D-D illustrated in FIG. 13 (a).

[0027] FIG. 14 is a sectional view illustrating a closed state where a lid body is placed on a discharge cap according to a second embodiment.

[0028] FIG. 15 is an enlarged view of a portion E illustrated in FIG. 14.

[0029] FIG. 16 is a sectional view of a discharge cap according to a modification of the second embodiment.

DESCRIPTION OF EMBODIMENTS

First Embodiment

[0030] Embodiments of the present invention will be described below in detail with reference to the drawings. Constituent elements, types, combinations, shapes, and relative arrangements described in the embodiments are merely explanatory examples, and are not intended to limit the scope of the present invention solely thereto unless otherwise specified.

[0031] FIGS. 1 and 2 are sectional views illustrating a discharge cap 1A and a part of a container 3 according to a first embodiment. As illustrated in FIGS. 1 and 2, the discharge cap 1A according to the first embodiment includes a discharge nozzle 29 and is removably attached to a mouth part 15 of the container (a squeeze container) 3 having an

elasticity. A liquid (a content liquid) CLq is stored in the interior space of the container 3. The discharge cap 1A discharges the content liquid CLq from the discharge nozzle 29 with a pressure increase in the interior space of the container 3 due to an external force applied to the container 3.

<Container 3>

[0032] The container 3 is molded of a synthetic resin such as polyethylene (PE) or polyethylene terephthalate (PET), and easily deforms and has an elasticity. The container 3 deforms so as to decrease the volume of the interior space when an external force is applied by compression (squeeze) or the like, and is restored to the original shape due to the elasticity when the external force is canceled.

[0033] The container 3 includes a body 13 for storing the content liquid CLq, and the mouth part 15 provided integrally with the body 13 and through which the content liquid CLq is taken in and out of the body 13. An external thread 17 for attaching the discharge cap 1A is formed on an outer circumferential portion of the mouth part 15.

[0034] The shape of the body 13 is not particularly limited thereto as long as the volume of the interior space is decreased when an external force is applied thereto and the body is restored to the original shape when the external force is canceled. For example, the body 13 can be formed in a bottomed cylindrical shape.

[0035] The content liquid CLq stored in the interior space of the container 3 is a viscous liquid such as a lubricant, a caulking agent, or honey. The viscosity of a lubricant is about 3100 mPa·s and the viscosity of honey is about 12000 mPa·s.

[0036] In the following descriptions, the content liquid CLq is, for example, a liquid having a viscosity of 500 mPa·s to 12000 mPa·s.

<Discharge Cap 1a According to First Embodiment>

[0037] The discharge cap 1A includes a cap body 5 that includes the discharge nozzle 29 and that is removably attached to the mouth part 15 of the container 3, a slit valve 7A (a valve body) and a holder 9 (a holding part) provided at a leading end part of the discharge nozzle 29, and a lid body 11 removably attached to the cap body 5 to cover the discharge nozzle 29.

[0038] The cap body 5 includes a ceiling wall 21 in a substantially discoid shape, a tubular outer wall part 23, a tubular intermediate wall part 25, and a tubular inner wall part 27, which extend from the inner surface of the ceiling wall 21 toward the container 3 and have a substantially cylindrical shape, and the discharge nozzle 29 in a substantially cylindrical shape extending from a central portion of the outer surface of the ceiling wall 21 to the opposite side to the container 3. In the present embodiment, the cap body 5 is made of a synthetic resin with such a hardness that does not deform at the time of discharge of the content liquid CLq.

[0039] The tubular outer wall part 23 extends from an outer edge part of the ceiling wall 21 to a region close to the body 13. An annular extended part 31 extended outward in the radial direction is formed on the outer wall surface of an end part of the tubular outer wall part 23 (an end part on the

side far from the ceiling wall 21). The extended part 31 is provided to support the opening edge of the lid body 11 attached to the cap body 5.

[0040] The tubular intermediate wall part 25 is provided on the side of the inner circumference of the tubular outer wall part 23 concentrically with the tubular outer wall part 23 at an appropriate distance therefrom. An end part of the tubular intermediate wall part 25 (an end part on the side far from the ceiling wall 21) extends close to the base of the mouth part 15. An internal thread 33 with which the external thread 17 of the mouth part 15 is removably fitted is formed on the inner wall surface of the tubular intermediate wall part 25.

[0041] The tubular inner wall part 27 is provided on the side of the inner circumference of the tubular intermediate wall part 25 concentrically with the tubular intermediate wall part 25 at a distance necessary for the leading end portion of the mouth part 15 to be fitted therewith. Therefore, a space (a gap) 35 in which the mouth part 15 of the container 3 is inserted is formed between the tubular intermediate wall part 25 and the tubular inner wall part 27.

[0042] The length of the tubular inner wall part 27 is smaller than that of the tubular intermediate wall part 25. A taper having the diameter decreasing with the increasing distance from the ceiling wall 21 is provided on the outer circumferential surface of the tubular inner wall part 27. The taper guides the mouth part 15 when the mouth part 15 is fitted to the cap body 5. An outer circumferential surface of the tubular inner wall part 27 on the side of the ceiling wall 21 tightly adheres to the inner circumferential surface of the mouth part 15 to suppress leakage of the content liquid CLq in a state where the mouth part 15 is fitted to the cap body 5.

<Discharge Nozzle 29>

[0043] The discharge nozzle 29 has a tubular shape extending from the base end on a side near the ceiling wall 21 toward the leading end far from the ceiling wall 21, and a nozzle flow path 29a through which the content liquid CLq flows is provided inside the discharge nozzle 29. The ceiling wall 21 is provided with an opening part 43 that causes the nozzle flow path 29a to be communicated with the interior space of the container 3. Therefore, the content liquid CLq flows so as to be capable of moving back and forth between the interior space of the container 3 and the nozzle flow path 29a.

[0044] A sectional area SA29a of the nozzle flow path 29a is smaller than an opening area OP15 of the mouth part 15. Accordingly, the thickness of the content liquid CLq discharged from the leading end of the discharge nozzle 29 can be thinned as compared to the content liquid CLq directly flowed out from the mouth part 15. Furthermore, since the discharge nozzle 29 has a predetermined length, the content liquid CLq can be easily discharged to a desired place.

[0045] In the present embodiment, the discharge nozzle 29 has a cylindrical shape tapered from the base end on the side near the ceiling wall 21 toward the leading end. However, the discharge nozzle 29 is not limited to this shape. For example, the sectional shape of the nozzle flow path 29a may be uniform from the base end of the discharge nozzle 29 toward the leading end.

[0046] FIG. 3 is an enlarged view of a portion A illustrated in FIG. 1. As illustrated in FIGS. 2 and 3, an annular protruding part 39 protruded inward in the radial direction is

formed on the inner circumference side of the leading end part of the discharge nozzle 29.

[0047] The inner circumferential surface of the protruding part 39 is a discharge opening 37 on which a valve part 49 and a thin-walled tube part 51 included in the slit valve 7A are arranged. The inner surface of the protruding part 39 (an annular face protruded inward in the radial direction in the nozzle flow path 29a) is an abutting face 38 that abuts on a valve sidewall part 47 of the slit valve 7A to prevent the slit valve 7A from falling off.

[0048] A diameter expanded part 45 with a diameter expanded outward in the radial direction is formed on the outer circumferential surface of the leading end part of the discharge nozzle 29. The diameter expanded part 45 is provided to removably fix the lid body 11.

[0049] As illustrated in FIG. 3, an annular concave part 41 is formed on the inner circumferential surface of the discharge nozzle 29 and at a position nearer the leading end in relation to the intermediate position of the discharge nozzle 29 in the length direction. The concave part 41 is provided to prevent the holder 9 from falling off the discharge nozzle 29.

<Slit Valve 7a>

[0050] FIG. 4 (a) is a plan view of the slit valve 7A and FIG. 4 (b) is a sectional view along a line B-B illustrated in FIG. 4 (a). As illustrated in FIGS. 3, 4 (a), and 4 (b), the slit valve 7A includes the cylindrical valve sidewall part 47 (a held part), the substantially discoid valve part 49 from which the content liquid CLq in the nozzle flow path 29a is discharged, and the thin-walled tube part 51 (a connecting part) that connects the valve sidewall part 47 and the valve part 49 to each other.

[0051] The slit valve 7A is made of an elastic material softer than the discharge nozzle 29 (the cap body 5). In the present embodiment, the slit valve 7A is integrally molded of an elastic synthetic resin such as silicone having a hardness (a hardness measured by a durometer of type A) of about A20° to about A50°. Since being made by integral molding, the slit valve 7A of the present embodiment is easily produced.

[0052] The valve sidewall part 47 is held on the discharge nozzle 29 by the holder 9. A leading end face 47a of the valve sidewall part 47 on a side near the thin-walled tube part 51 (more specifically, a face projecting outward in the radial direction relative to the thin-walled tube part 51) abuts on the abutting face 38 of the protruding part 39 to prevent the slit valve 7A from falling off the discharge nozzle 29. Accordingly, a thickness TH47 of the valve sidewall part 47 is determined to a thickness required to prevent falling off the discharge nozzle 29. In the present embodiment, the thickness TH47 of the valve sidewall part 47 is about 0.3 mm (millimeter) to about 1.0 mm.

[0053] The thin-walled tube part 51 connects an inner circumferential part of the valve sidewall part 47 on a side near the leading end face 47a and an outer circumferential part of the valve part 49 on a side near an inner surface 54 to each other. The outside diameter of the thin-walled tube part 51 is substantially equal to the opening diameter of the discharge opening 37. Accordingly, the outer circumferential surface of the thin-walled tube part 51 abuts on the discharge opening 37 in a state where the slit valve 7A is held on the leading end part of the discharge nozzle 29. In other words, the thin-walled tube part 51 is arranged so as to abut on the discharge opening 37 of the discharge nozzle 29.

[0054] A thickness TH51 of the thin-walled tube part 51 is sufficiently smaller than the thickness TH47 of the valve sidewall part 47. For example, the thickness TH51 of the thin-walled tube part 51 is equal to or less than a half (substantially a quarter to substantially a third) of the thickness TH47 of the valve sidewall part 47. A length L51 of the thin-walled tube part 51 is about 0.3 mm to about 0.7 mm.

[0055] Accordingly, the area of the leading end face 47a of the valve sidewall part 47 (in other words, the area of a portion abutting on the abutting face 38 of the discharge nozzle 29) can be sufficiently provided and falling of the slit valve 7A from the discharge nozzle 29 can be effectively prevented.

[0056] The thin-walled tube part 51 more easily deforms than the valve sidewall part 47. Even when the valve sidewall part 47 deforms associated with attachment of the slit valve 7A, the thin-walled tube part 51 easily deforms and accordingly absorbs the deformation of the valve sidewall part 47.

[0057] To specifically explain this configuration, the valve sidewall part 47 is sandwiched by the discharge nozzle 29 and the holder 9 from the outer circumference side and the inner circumference side as illustrated in FIG. 3. Attachment of the slit valve 7A to the discharge nozzle 29 is performed, for example, by relatively moving the holder 9 having the slit valve 7A put on the leading end part toward the leading end of the discharge nozzle 29 through an inner void (that is, the nozzle flow path 29a) of the discharge nozzle 29. Therefore, at the time of attachment of the slit valve 7A, the valve sidewall part 47 is sometimes deformed by friction between the inner circumferential surface of the discharge nozzle 29 and the outer circumferential surface of the holder 9. Even when the valve sidewall part 47 is deformed, the thin-walled tube part 51 deforms itself to prevent the valve part 49 from being affected by the deformation.

[0058] There is also a case where the thin-walled tube part 51 deforms at the time of discharge of the content liquid CLq. For example, in a case where the content liquid CLq having a relatively high viscosity, such as a lubricant (viscosity: about 3100 mPa·s) or honey (viscosity: about 12000 mPa·s) is discharged, the thin-walled tube part 51 extends when an external force in the discharge direction is applied to the valve part 49 and contracts after the content liquid CLq is discharged. As will be described later, the thin-walled tube part 51 pulls an outer circumferential part 49a toward the nozzle flow path 29a at the time of contraction, and accordingly assists deformation of a slit part 49b toward the nozzle flow path 29a. As a result, the content liquid CLq after discharge is cut off in an improved manner.

<Valve Part 49>

[0059] The outside diameter of the valve part 49 is substantially equal to the opening diameter of the discharge opening 37. Accordingly, the outer circumferential surface of the valve part 49 is in contact with the discharge opening 37 in a state where the slit valve 7A is held on the leading end part of the discharge nozzle 29. In other words, the valve part 49 is arranged so as to be fitted in the discharge opening 37 of the discharge nozzle 29. In the present embodiment, the outside diameter 049 of the valve part 49 is about 4 mm to about 7 mm.

[0060] An outer surface 53 of the valve part 49 is a concave face spherically depressed toward the side of the

nozzle flow path 29a (on the opposite side to the discharge direction of the content liquid CLq). Meanwhile, the inner surface 54 of the valve part 49 is a flat face orthogonal to a central axis line L of the cylinder of the valve sidewall part 47.

[0061] Slits 55 extending outward in the radial direction from a center point O are formed on the valve part 49. In the slit valve 7A, the slits 55 are cross-shaped. The slits 55 are configured to open and close according to a difference between the pressure in the interior space of the container 3 and the atmospheric pressure. In the present embodiment, a length L55 of each of the slits 55 is about 2 mm to about 5 mm.

[0062] As for the slits 55, the intersecting angle between one slit and the other slit is not limited to 90 degrees. For example, the intersecting angle between one slit and the other slit may be changed in a range from 60 degrees to 90 degrees. The length of one slit and the length of the other slit may be different.

[0063] The valve part 49 includes the outer circumferential part 49a coupled to the thin-walled tube part 51, a slit part 49b on which the slits 55 are provided, and an intermediate part 49c located between the slit part 49b and the outer circumferential part 49a.

[0064] As described above, the outer surface 53 of the valve part 49 is a concave face in a spherical shape and the inner surface 54 is a flat face. Therefore, the outer circumferential part 49a, the intermediate part 49c, and the slit part 49b are formed to be gradually thinned from the outer circumference of the valve part 49 toward the center point O. Accordingly, the intermediate part 49c is formed to be gradually thinned from the outer circumferential part 49a toward the slit part 49b.

[0065] In the present embodiment, the thickness of the slit part 49b is equal to or less than a half of the thickness of the outer circumferential part 49a and the slit part 49b is provided on a side nearer the nozzle flow path 29a in relation to an intermediate position TC49 of the outer circumferential part 49a in the thickness direction.

[0066] To cite an example, a largest thickness TH49a of the outer circumferential part 49a is about 0.6 mm to about 0.7 mm and a largest thickness TH49b of the slit part 49b is about 0.3 mm to about 0.35 mm. A smallest thickness of the slit part 49b at the center point O is about 0.15 mm to about 0.25 mm.

[0067] With the configuration described above, at the time of discharge of the content liquid CLq, the slit part 49b more easily deforms than the outer circumferential part 49a due to a pressure change of the content liquid CLq (the interior space of the container 3) and contributes to reduction of the diameter of the leading end side (downstream side) of the discharge nozzle 29.

[0068] The outer circumferential part 49a is thicker than the intermediate part 49c and the slit part 49b, and the intermediate part 49c is thicker than the slit part 49b. Accordingly, after discharge of the content liquid CLq, the outer circumferential part 49a has a larger force (restoring force) to deform in a returning direction than the intermediate part 49c, and the intermediate part 49c has a larger restoring force than the slit part 49b. As a result, the slit part 49b is promptly deformed toward the nozzle flow path 29a under the restoring forces from the outer circumferential part

49a and the intermediate part 49c in addition to its own restoring force, and improves cutoff of the content liquid CLq.

[0069] Furthermore, since the outer circumferential part 49a and the intermediate part 49c are thicker than the slit part 49b, the both ends of each of the slits 55 are enabled to become less likely to be split.

[0070] Besides, since the slit valve 7A is integrally molded of a synthetic resin such as silicone, the softness of the parts 49a, 49b, and 51 can be easily set by setting the thicknesses TH49a, TH49b, and TH51 of the parts 49a, 49b, and 51.

<Holder 9>

[0071] As illustrated in FIG. 3, the holder 9 is a hollow tubular body fitted in the inner circumferential part of the discharge nozzle 29 on the leading end side and is a substantially cylindrical member in the present embodiment. The holder 9 holds the valve sidewall part 47 along with the discharge nozzle 29 and prevents the slit valve 7A from falling off the discharge nozzle 29.

[0072] The holder 9 includes a small-diameter tube part 57 located on the side of the leading end of the discharge nozzle 29, and a large-diameter tube part 59 located on a side nearer the base end of the discharge nozzle 29 in relation to the small-diameter tube part 57. The base end of the small-diameter tube part 57 and the leading end of the large-diameter tube part 59 is integrally coupled to each other. An interior space 9a of the holder 9 constitutes a part of the nozzle flow path 29a.

[0073] The outside diameter of the small-diameter tube part 57 is substantially the same as the inside diameter of the valve sidewall part 47. The outside diameter of the large-diameter tube part 59 is substantially the same as the inside diameter of the discharge nozzle 29.

[0074] A flange part 61 protruded outward in the radial direction is provided on a base end portion of the outer circumferential surface of the large-diameter tube part 59. The flange part 61 is fitted in the annular concave part 41 of the discharge nozzle 29.

[0075] The holder 9 is positioned and is fixed to the discharge nozzle 29 by fitting of the flange part 61 into the concave part 41.

[0076] When the holder 9 is fixed to the discharge nozzle 29, the valve sidewall part 47 of the slit valve 7A is arranged between the discharge nozzle 29 and the small-diameter tube part 57.

[0077] The leading end face 47a of the valve sidewall part 47 abuts on the abutting face 38 of the protruding part 39, and a base end face 47b of the valve sidewall part 47 abuts on a leading end face 60 (a face with a diameter expanded outward in the radial direction relative to the small-diameter tube part 57) of the large-diameter tube part 59.

[0078] As described above, the valve sidewall part 47 is held in a space formed by the discharge nozzle 29 and the holder 9. As a result, falling of the slit valve 7A from the discharge nozzle 29 is prevented. The thin-walled tube part 51 extends from a gap between the small-diameter tube part 57 and the protruding part 39 to the side of the leading end of the discharge nozzle 29.

<Lid Body 11>

[0079] As illustrated in FIGS. 1 and 3, the lid body 11 is a member having a bottomed cylindrical shape including a

cylindrical wall part 65 in a cylindrical shape, and a top board 67 in a discoid shape. A cylindrical latch part 69 is provided in a protruded manner at a central portion of an inner surface 67a of the top board 67.

[0080] The leading end portion of the discharge nozzle 29 is inserted into the interior space of the latch part 69. On the inner circumferential surface of an open end part of the latch part 69, a taper expanding toward the open end is formed. This taper is provided to guide the leading end portion of the discharge nozzle 29.

[0081] An abutting part 71 raised in a spherical shape toward the open end of the latch part 69 is provided at a portion of the top board 67 on an inner circumference side than the latch part 69. The surface of the abutting part 71 is brought to contact with the outer surface 53 of the valve part 49 to enhance the sealing property of the valve part 49 at the time of attachment of the lid body 11 to the cap body 5.

<Operation at Time of Discharging Content Liquid CLq>

[0082] An operation at the time of discharging a content liquid CLq is explained next.

[0083] FIG. 5(a) is a view illustrating a state where the nozzle flow path 29a is filled with the content liquid CLq and no external force is applied to the container 3, FIG. 5(b) is a view illustrating a state where the slit part 49b of the slit valve 7A is pushed and spread by the content liquid CLq in the nozzle flow path 29a, and FIG. 5(c) is a view illustrating a state where the content liquid CLq is discharged outside from the slit part 49b.

[0084] FIGS. 5(a) to 5(c) illustrate the leading end part of the discharge nozzle 29. In these drawings, the leading end part of the discharge nozzle 29 faces downward.

[0085] An operation in a case where an external force is applied to the container 3 is explained first. As illustrated in FIG. 5(a), the slit part 49b of the slit valve 7A is in a closed state in a state where the nozzle flow path 29a is filled with the content liquid CLq and no external force is applied to the container 3.

[0086] Therefore, the content liquid CLq in the nozzle flow path 29a is kept in the nozzle flow path 29a by the slit valve 7A (the valve part 49) in a closed state.

[0087] As illustrated in FIG. 5(b), when an external force is applied to the container 3 by compression or the like, the pressure in the interior space is increased and a force PR+ in the discharge direction is applied to the content liquid CLq in the nozzle flow path 29a. The force PR+ in the discharge direction is proportional to a difference between the pressure in the interior space of the container 3 and the atmospheric pressure. The force PR+ in the discharge direction is transmitted through the content liquid CLq and acts on the inner surface 54 of the valve part 49.

[0088] The slit part 49b is deformed in the discharge direction by the content liquid CLq. With the deformation of the slit part 49b, a gap (a discharge port) is formed between the slits 55.

[0089] Since being smaller in the thickness than the outer circumferential part 49a and the intermediate part 49c, the slit part 49b deforms more largely than the outer circumferential part 49a and the intermediate part 49c. Furthermore, since being smaller in the thickness than the outer circumferential part 49a, the intermediate part 49c deforms more largely than the outer circumferential part 49a. Accordingly, the inner surface 54 of the valve part 49 forms

an inclined face toward the discharge port and smoothly guides the content liquid CLq.

[0090] Subsequently, as illustrated in FIG. 5 (c), the content liquid CLq is continuously discharged from the gap (the discharge port) between the slits 55. In this discharge cap 1A, the thickness of the slit part 49b is smaller than those of the outer circumferential part 49a and the intermediate part 49c. Accordingly, the slit part 49b deforms more largely with a pressure change of the content liquid CLq (a pressure change in the interior space of the container 3) than the outer circumferential part 49a and the intermediate part 49c.

[0091] As a result, the content liquid CLq can be discharged from the discharge nozzle 29 without application of an excessive external force to the container 3. Furthermore, since the thickness of the content liquid CLq discharged from the discharge nozzle 29 can be limited by the size of the slit part 49b (the slits 55), an inconvenience that the content liquid CLq becomes too thick is suppressed.

[0092] An operation at the time of sucking the content liquid CLq into the discharge nozzle 29 is explained next.

[0093] FIG. 6 (a) is a view illustrating a state where the content liquid CLq in the nozzle flow path 29a starts returning to the container 3 in a state where the external force to the container 3 is canceled, FIG. 6 (b) is a view illustrating a state where air AR starts flowing into the nozzle flow path 29a, and FIG. 6 (c) is a view illustrating a state where the flow of the air AR into the nozzle flow path 29a ends and the slit valve 7A is closed.

[0094] Also in FIGS. 6 (a) to 6 (c), the leading end part of the discharge nozzle 29 faces downward similarly in FIGS. 5 (a) to 5 (c).

[0095] As illustrated in FIG. 6 (a), when the external force to the container 3 is canceled after discharge of the content liquid CLq, a force PR- (hereinafter, referred to as “force PR- in the returning direction”) to return to the container 3 is applied to the content liquid CLq in the nozzle flow path 29a. The force PR- in the returning direction is generated because the container 3 in the contracted state is expanding to the original shape with its elasticity.

[0096] Besides, the outer circumferential part 49a, the intermediate part 49c, and the slit part 49b are also restoring to the original shapes with their elasticities. Since being larger in the thickness than the intermediate part 49c and the slit part 49b, the outer circumferential part 49a has a larger restoring force than the intermediate part 49c and the slit part 49b. Similarly, the intermediate part 49c has a larger restoring force than the slit part 49b.

[0097] As a result, after discharge of the content liquid CLq, the slit part 49b is promptly deformed toward the nozzle flow path 29a under the restoring forces of the outer circumferential part 49a and the intermediate part 49c in addition to its own restoring force.

[0098] As illustrated in FIG. 6 (b), after the content liquid CLq passes through the valve part 49, an amount of air AR compensating the discharged content liquid CLq flows in the nozzle flow path 29a. At the time of inflow of the air AR, the slit part 49b deforms toward the nozzle flow path 29a to form a gap between the slits 55.

[0099] The thickness of the slit part 49b is smaller than those of the outer circumferential part 49a and the intermediate part 49c. Accordingly, the slit part 49b deforms more largely than the outer circumferential part 49a and the intermediate part 49c. With the deformation of the slit part

49b, the gap between the slits 55 can have a sufficient size and the air AR can be efficiently taken into the nozzle flow path 29a.

[0100] The thin-walled tube part 51 formed to have a thin wall is sometimes inclined in a circumferentially inward direction of the thin-walled tube part 51. For example, the thin-walled tube part 51 is sometimes inclined in the circumferentially inward direction when the amount of air AR taken into the nozzle flow path 29a is sufficiently large, when the speed at which the air AR is taken in is sufficiently high, or when the viscosity of the content liquid CLq is high. When the thin-walled tube part 51 is inclined, the slit part 49b is largely inclined toward the nozzle flow path 29a correspondingly to the inclination of the thin-walled tube part 51. Accordingly, the gap between the slits 55 becomes larger than in a case where the thin-walled tube part 51 is not inclined. As a result, also with the inclination of the thin-walled tube part 51, the air AR can be efficiently taken into the nozzle flow path 29a.

[0101] Subsequently, as illustrated in FIG. 6 (c), the force PR- in the returning direction to the content liquid CLq is lost with restoration of the container 3 to the original shape. With the loss of the force PR- in the returning direction, the slit part 49b of the slit valve 7A is brought to the closed state.

[0102] Therefore, the content liquid CLq in the nozzle flow path 29a is kept in the nozzle flow path 29a by the slit valve 7A (the valve part 49) in a closed state.

[0103] As is apparent from the above descriptions, the slit part 49b deforms according to a pressure change of the content liquid CLq (a pressure change in the interior space of the container 3). Therefore, the gap between the slits 55 can be made sufficiently large by the force PR- in the returning direction. Accordingly, the gap is enabled to have a size required to efficiently suck the content liquid CLq and air AR into the nozzle flow path 29a, and the content liquid CLq can be cut off in an improved manner.

[0104] In the slit valve 7A, since the outer circumferential part 49a can be formed to have a sufficient thickness, the restoring force of the outer circumferential part 49a after discharge of the content liquid CLq can be enhanced. Similarly, the restoring force of the intermediate part 49c also can be enhanced as compared to the slit part 49b.

[0105] The force PR+ in the discharge direction applied to the valve part 49 (particularly, to the inner surface 54 of the valve part 49) at the time of discharge of the content liquid CLq is increased as the viscosity of the content liquid CLq is increased. Since being thinner and more easily deforming than the outer circumferential part 49a of the valve part 49, the thin-walled tube part 51 is sometimes extended due to the force PR+ in the discharge direction applied to the outer circumferential part 49a as indicated by an arrow in FIG. 5 (b).

[0106] When the external force to the container 3 is canceled while the thin-walled tube part 51 remains extended, the thin-walled tube part 51 starts contracting in the return direction with its elasticity as illustrated by an arrow in FIG. 6 (a). The outer circumferential part 49a is pulled toward the container 3 by contraction of the thin-walled tube part 51. The contractive force of the thin-walled tube part 51 assists deformation of the slit part 49b toward the nozzle flow path 29a after discharge of the content liquid CLq.

[0107] As described above, there is a case where the content liquid CLq is enabled to be cut off in a further improved manner also by extension and contraction of the thin-walled tube part 51.

[0108] The discharge nozzle 29 of the discharge cap 1A has a cylindrical shape tapered from the base end (the upstream side) toward the leading end (the downstream side). Therefore, the content liquid can be easily discharged into a recess. As a result, the content liquid CLq can be easily discharged to various places.

<Variations of Slit Valve>

[0109] Variations of the slit valve are explained below.

[0110] FIG. 7 (a) is a plan view of a slit valve 7B according to a first variation, FIG. 7 (b) is a sectional view along a line C-C illustrated in FIG. 7 (a), FIG. 8 is a sectional view of a slit valve 7C according to a second variation, FIG. 9 is a sectional view of a slit valve 7D according to a third variation, FIG. 10 is a sectional view of a slit valve 7E according to a fourth variation, and FIG. 11 is a sectional view of a slit valve 7F according to a fifth variation.

<Slit Valve 7b According to First Variation>

[0111] The slit valve 7B according to the first variation illustrated in FIGS. 7 (a) and 7 (b) is characterized in that a slit 55B has a straight line shape, in other words, the slit 55B is formed of one linear slit.

[0112] While being different from the slit valve 7A described above in that the slit 55B is formed of one linear slit, the slit valve 7B is the same as the slit valve 7A in the rest of the configuration.

[0113] Therefore, also with use of the slit valve 7B in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0114] In the slit valve 7B, the thickness of the slit part 49b is equal to or less than a half of the thickness of the outer circumferential part 49a and the slit part 49b is provided on a side nearer the nozzle flow path 29a in relation to the intermediate position TC49 of the outer circumferential part 49a in the thickness direction.

[0115] Therefore, also with use of the slit valve 7B in the discharge cap 1A, the restoring force of the outer circumferential part 49a after discharge of the content liquid CLq can be enhanced.

[0116] With the configuration described above, the slit part 49b is promptly deformed toward the nozzle flow path 29a after discharge of the content liquid CLq. As a result, the content liquid CLq can be cut off in an improved manner. Also in the slit valve 7B, when the viscosity of the content liquid CLq is sufficiently high, the outer circumferential part 49a is pulled toward the container 3 by contraction of the thin-walled tube part 51 and accordingly the content liquid CLq can be cut off in an improved manner.

<Slit valve 7C according to second variation>

[0117] The slit valve 7C according to the second variation illustrated in FIG. 8 is characterized in that an outer surface 53C of the valve part 49 is a flat face and that an inner

surface 54C is a concave face spherically depressed in the discharge direction of the content liquid CLq.

[0118] Also with use of the slit valve 7C in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0119] The outer circumferential part 49a and the intermediate part 49c promptly deform the slit part 49b toward the nozzle flow path 29a after discharge of the content liquid CLq. As a result, the content liquid CLq can be cut off in an improved manner. Furthermore, when the viscosity of the content liquid CLq is sufficiently high, the content liquid CLq can be cut off in a further improved manner by extension and contraction of the thin-walled tube part 51.

<Slit Valve 7d According to Third Variation>

[0120] The slit valve 7D according to the third variation illustrated in FIG. 9 is characterized in that each of an outer surface 53D and an inner surface 54D of the valve part 49 is a concave face spherically depressed.

[0121] Also with use of the slit valve 7D in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0122] The outer circumferential part 49a and the intermediate part 49c promptly deform the slit part 49b toward the nozzle flow path 29a after discharge of the content liquid CLq. As a result, the content liquid CLq can be cut off in an improved manner. Furthermore, when the viscosity of the content liquid CLq is sufficiently high, the content liquid CLq can be cut off in a further improved manner by extension and contraction of the thin-walled tube part 51.

<Slit Valve 7b According to Fourth Variation>

[0123] The slit valve 7E according to the fourth variation illustrated in FIG. 10 is characterized in that the thickness of the intermediate part 49c of the valve part 49 is stepwise thinned toward the center of the valve part 49 and that the slit part 49b is provided on a side nearer the nozzle flow path 29a in relation to the intermediate position TC49 of the outer circumferential part 49a in the thickness direction.

[0124] Also with use of the slit valve 7E in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0125] Furthermore, the thickness of the slit part 49b is equal to or less than a half of the thickness of the outer circumferential part 49a and the slit part 49b is provided on the side nearer the nozzle flow path 29a in relation to the intermediate position TC49. Accordingly, after discharge of the content liquid CLq, the slit part 49b is promptly deformed toward the nozzle flow path 29a. As a result, the content liquid CLq can be cut off in an improved manner after discharge of the content liquid CLq.

<Slit Valve 7F According to Fifth Variation>

[0126] The slit valve 7F according to the fifth variation illustrated in FIG. 11 is characterized in that a portion of an outer surface 53F of the valve part 49 corresponding to the intermediate part 49c and a portion of an inner surface 54F corresponding to the intermediate part 49c are tapered faces inclined toward the slit part 49b, and that the slit part 49b is provided on a side nearer the nozzle flow path 29a in relation to the intermediate position TC49 of the outer circumferential part 49a in the thickness direction.

[0127] Also with use of the slit valve 7F in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0128] Furthermore, the thickness of the slit part 49b is equal to or less than a half of the thickness of the outer circumferential part 49a and the slit part 49b is provided on the side nearer the nozzle flow path 29a in relation to the intermediate position TC49. Accordingly, after discharge of the content liquid CLq, the slit part 49b is promptly deformed toward the nozzle flow path 29a. As a result, the content liquid CLq can be cut off in an improved manner after discharge of the content liquid CLq.

<Discharge Cap 1B According to Modification of First Embodiment>

[0129] FIG. 12 is a sectional view of a discharge cap 1B according to a modification of the first embodiment, FIG. 13 (a) is a plan view illustrating the discharge cap 1B according to the modification of the first embodiment in an open state, and FIG. 13 (b) is a sectional view along a line D-D illustrated in FIG. 13 (a).

[0130] As illustrated in FIGS. 12, 13 (a), and 13 (b), the discharge cap 1B according to the modification of the first embodiment is characterized in that the discharge nozzle 29 is shorter than the discharge cap 1A of the first embodiment. Configurations identical to those of the discharge cap 1A according to the first embodiment are denoted by like reference signs and explanations thereof are omitted.

[0131] The lid body 11 is attached to the cap body 5 of the discharge cap 1B with a hinge 75. A projection 79 is provided on the lid body 11 on the opposite side to the hinge 75. A concave part 77 is provided on the cap body 5 on the opposite side to the hinge 75. The projection 79 is provided to be caught by a finger or a nail of a user in a closed state (see FIG. 12) where the lid body 11 is placed on the cap body 5. The concave part 77 is provided to enable a finger or the like of a user to easily catch the projection 79.

[0132] As described above, also in the discharge cap 1B according to the modification having the shorter discharge nozzle 29, the content liquid CLq can be discharged from the discharge nozzle 29 without application of an excessively large external force to the container 3. Also in the discharge cap 1B, the diameter of the leading end part of the discharge nozzle 29 can be decreased to enable a small amount of the content liquid CLq to be discharged.

<Discharge Cap 1C According to Second Embodiment>

[0133] FIG. 14 is a sectional view illustrating a closed state where the lid body 11 is placed on a discharge cap 1C

according to a second embodiment, and FIG. 15 is an enlarged view of a portion E illustrated in FIG. 14.

[0134] As illustrated in FIGS. 14 and 15, the discharge cap 1C according to the second embodiment is characterized in that a holder 9C is attached to the outer circumferential surface of the leading end part of a discharge nozzle 29C as compared to the discharge cap 1A according to the first embodiment. Configurations identical to those of the discharge cap 1A according to the first embodiment are denoted by like reference signs and explanations thereof are omitted.

[0135] The discharge nozzle 29C includes a large-diameter cylindrical wall 81 having the largest diameter, an intermediate-diameter cylindrical wall 83 smaller in the diameter than the large-diameter cylindrical wall 81, and a small-diameter cylindrical wall 85 smaller in the diameter than the intermediate-diameter cylindrical wall 83 arranged in this order from the side of the ceiling wall 21 toward the leading end side. In the present embodiment, the large-diameter cylindrical wall 81 is the longest, and accounts for about three-fourths of the length of the discharge nozzle 29C. The intermediate-diameter cylindrical wall 83 is the second longest and the small-diameter cylindrical wall 85 is the shortest. Therefore, it can be said that the discharge nozzle 29C is stepwise reduced in the outside diameter at a portion on the leading end side.

[0136] The outside diameter of the small-diameter cylindrical wall 85 is substantially the same as the inside diameter of the valve sidewall part 47 of the slit valve 7A. An annular projecting part 87 is formed on the outer circumferential surface of the intermediate-diameter cylindrical wall 83. The projecting part 87 is provided to prevent falling of the holder 9C.

[0137] The holder 9C is a substantially cylindrical member and the inside diameter of the holder 9C is configured to be substantially the same as the outside diameter of the intermediate-diameter cylindrical wall 83 except for a leading end portion thereof. In the present embodiment, the outside diameter of the intermediate-diameter cylindrical wall 83 is configured to be substantially the same as the outside diameter of the cylindrical valve sidewall part 47 of the slit valve 7A. The holder 9C is fitted with the intermediate-diameter cylindrical wall 83 of the discharge nozzle 29C. In a state where the holder 9C is fitted therewith, the small-diameter cylindrical wall 85 is arranged in the interior space of the holder 9C.

[0138] On the inner circumference side of the leading end part of the holder 9C, an annular protruding part 89 protruded inward in the radial direction is formed. The protruding part 89 is provided for the same purpose as that of the protruding part 39 included in the discharge cap 1A of the first embodiment. A diameter expanded part 93 that engages with the latch part 69 of the lid body 11 is provided on the outer circumferential surface of the leading end part of the holder 9C.

[0139] An annular concave part 91 is formed on the inner circumferential surface of the holder 9C at a position near the base end. The concave part 91 fits with the projecting part 87 provided on the outer circumferential surface of the intermediate-diameter cylindrical wall 83 and prevents falling of the holder 9C from the discharge nozzle 29C.

[0140] According to the configuration described above, in the discharge cap 1C, attachment of the holder 9C to the discharge nozzle 29C enables the cylindrical valve sidewall part 47 of the slit valve 7A to be held between the inner

circumferential surface of the holder 9C and the outer circumferential surface of the small-diameter cylindrical wall 85 of the discharge nozzle 29C, and enables the valve part 49 to be held by the inner circumferential portion of the annular protruding part 89.

[0141] The discharge cap 1C according to the second embodiment also exerts identical operational effects as those of the discharge cap 1A according to the first embodiment.

<Discharge Cap 1D According to Modification of Second Embodiment>

[0142] FIG. 16 is a sectional view of a discharge cap 1D according to a modification of the second embodiment. As illustrated in FIG. 16, the discharge cap 1D according to the modification of the second embodiment is obtained by applying the configurations of the discharge nozzle 29C and the holder 9C of the second embodiment to the discharge cap 1B (see FIG. 12 and the like) according to the modification of the first embodiment. Configurations identical to those of the discharge cap 1B according to the modification of the first embodiment and the discharge cap 1C according to the second embodiment are denoted by like reference signs and explanations thereof are omitted.

[0143] The discharge cap 1D according to the modification of the second embodiment also exerts identical operational effects as those of the discharge cap 1A according to the first embodiment.

<Other Modifications>

[0144] While the discharge caps 1A and 1C described above have a configuration in which the lid body 11 is fitted with the cap body 5, the discharge caps 1A and 1C are not limited to this configuration. For example, it is possible to form an external thread on the outer circumferential surface of the tubular outer wall part 23 of the cap body 5 and to form an internal thread on the inner circumferential surface of the cylindrical wall part 65 of the lid body 11.

[0145] While the discharge caps 1B and 1D described above have a configuration in which the lid body 11 and the cap body 5 are connected to each other with the hinge 75, the discharge caps 1B and 1C are not limited to this configuration. For example, a configuration in which the lid body 11 is fitted with the cap body 5 without connecting the lid body 11 and the cap body 5 to each other may be adopted.

[0146] In the discharge caps 1A, 1B, 1C, and 1D described above, the lengths of the discharge nozzles 29 and 29A can be appropriately determined.

Summary of Aspect Examples, Actions, and Effects of Present Invention

<First Aspect>

[0147] The present aspect provides the discharge cap 1A, 1B, 1C, or 1D that is attached to the mouth part 15 of the container 3 having an elasticity and that discharges a content liquid CLq associated with a pressure increase in an interior space of the container 3 due to an external force applied to the container 3, the discharge cap comprising: the discharge nozzle 29 or 29C that forms the nozzle flow path 29a with the sectional area SA29a smaller than the opening area OP15 of the mouth part 15; and the slit valve 7A, 7B, 7C, 7D, 7E, or 7F that is provided on the discharge nozzle 29 or 29C and that is made of an elastic material softer than the

discharge nozzle 29 or 29C, in which the slit valve 7A, 7B, 7C, 7D, 7E, or 7F includes the valve part 49 through which the content liquid CLq in the nozzle flow path 29a is discharged, the tubular valve sidewall part 47 (held part) that is held on the discharge nozzle 29 or 29C by the holder 9 or 9C, and the tubular thin-walled tube part 51 (a connecting part) that connects the valve part 49 and the valve sidewall part 47 to each other, the valve part 49 includes the slit part 49b provided with the slit 55 or 55B opening and closing according to a difference between an atmospheric pressure and a pressure in the interior space of the container 3, and the outer circumferential part 49a coupled to the thin-walled tube part 51, and the slit part 49b more easily deforms with a pressure change in the interior space than the outer circumferential part 49a, deforms in a discharge direction of the content liquid CLq at a time of discharge of the content liquid CLq, and deforms toward the nozzle flow path 29a after the content liquid CLq is discharged.

[0148] With the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the slit part 49b deforms with a pressure change in the interior space, and the slits 55 or the slit 55B opens to form a discharge port for the content liquid CLq. Since the slit part 49b more easily deforms than the outer circumferential part 49a, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and an inconvenience that the content liquid CLq discharged from the discharge nozzle 29 or 29C becomes too thick is suppressed.

[0149] Furthermore, since the outer circumferential part 49a is less likely to deform than the slit part 49b, the restoring force thereof after the deformation is larger than that of the slit part 49b. Accordingly, after discharge of the content liquid CLq, the restoring force of the outer circumferential part 49a can be increased. The restoring force of the outer circumferential part 49a assists the restoring force of the slit part 49b to return to the closed state. As a result, the content liquid CLq after discharge can be cut off in an improved manner.

<Second Aspect>

[0150] In the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the slit part 49b of the slit valve 7A, 7B, 7C, 7D, 7E, or 7F is integrally molded of a same elastic material as that of the outer circumferential part 49a, and the thickness TH49b of the slit part 49b and the thickness TH51 of the thin-walled tube part 51 are smaller than the thickness TH49a of the outer circumferential part 49a.

[0151] With the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the slit valve 7A, 7B, 7C, 7D, 7E, or 7F is easily produced and softness thereof can be managed by the thicknesses TH49a and TH49b of the parts 49a and 49b.

<Third Aspect>

[0152] In the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the valve part 49 of the slit valve 7A, 7B, 7C, 7D, 7E, or 7F includes the intermediate part 49c between the slit part 49b and the outer circumferential part 49a, and the intermediate part 49c is gradually thinned from the outer circumferential part 49a to the slit part 49b.

[0153] With the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, after discharge of the content liquid CLq, the outer circumferential part 49a has a larger restoring

force than the intermediate part 49c and the intermediate part 49c has a larger restoring force than the slit part 49b. As a result, the slit part 49b promptly deforms toward the nozzle flow path 29a under the restoring forces from the outer circumferential part 49a and the intermediate part 49c in addition to its own restoring force, and cuts off the content liquid CLq in an improved manner.

<Fourth Aspect>

[0154] In the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the valve part 49 of the slit valve 7A or 7B has the outer surface 53 being a spherical concave face depressed toward the nozzle flow path 29a, and the inner surface 54 being a flat face.

[0155] With the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the amount of deformation of the outer circumferential part 49a caused by a pressure from the content liquid CLq applied to the inner surface 54 of the outer circumferential part 49a can be increased. Accordingly, after discharge of the content liquid CLq, a force of the outer circumferential part 49a to deform in the returning direction can be increased and the slit part 49b can be promptly deformed toward the nozzle flow path 29a. As a result, the content liquid CLq can be cut off in an improved manner.

<Fifth Aspect>

[0156] In the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the slit part 49b in the valve part 49 of the slit valve 7A, 7B, 7E, or 7F has a thickness equal to or less than a half of a thickness of the outer circumferential part 49a, and the slit part 49b is located on a side nearer the nozzle flow path 29a in relation to the intermediate position TC49 of the outer circumferential part 49a in a thickness direction.

[0157] With the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the slit part 49b is provided to be shifted toward the nozzle flow path 29a in relation to the intermediate position TC49. Therefore, as compared to a configuration in which the slit part 49b is provided to be shifted toward the discharge side in relation to the intermediate position TC49, the amount of deformation of the slit part 49b at the time of discharge of the content liquid CLq can be increased. Therefore, the restoring force of the slit part 49b after discharge of the content liquid CLq is enhanced. As a result, the slit part 49b can be promptly deformed toward the nozzle flow path 29a and can cut off the content liquid CLq in an improved manner.

<Sixth Aspect>

[0158] In the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the thin-walled tube part 51 (the connecting part) of the slit valve 7A, 7B, 7C, 7D, 7E, or 7F more easily deforms with a pressure change in the interior space of the container 3 than the outer circumferential part 49a, extends in the discharge direction at a time of discharge of the content liquid CLq, and contracts in a returning direction after the content liquid CLq is discharged.

[0159] With the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, after the content liquid CLq is discharged, the outer circumferential part 49a can be pulled toward the nozzle flow path 29a with a contractive force of the thin-walled tube part 51. Therefore, the slit part 49b can

be promptly deformed toward the nozzle flow path 29a. As a result, the content liquid CLq can be cut off in an improved manner.

<Seventh Aspect>

[0160] In the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the thin-walled tube part 51 (the connecting part) of the slit valve 7A, 7B, 7C, 7D, 7E, or 7F more easily deforms with a pressure change in the interior space of the container 3 than the outer circumferential part 49a, and is inclined in a circumferentially inward direction of the thin-walled tube part 51 after the content liquid CLq is discharged.

[0161] With the discharge cap 1A, 1B, 1C, or 1D according to the present aspect, the gap between the slits 55 can be enlarged after discharge of the content liquid CLq, and air AR can be efficiently drawn into the nozzle flow path 29a.

REFERENCE SIGNS LIST

[0162] 1A . . . discharge cap according to first embodiment, 1B . . . discharge cap according to modification of first embodiment, 1C . . . discharge cap according to second embodiment, 1D . . . discharge cap according to modification of second embodiment, 3 . . . container, 5 . . . cap body, 7A . . . slit valve, 7B . . . slit valve according to first variation, 7C . . . slit valve according to second variation, 7D . . . slit valve according to third variation, 7E . . . slit valve according to fourth variation, 7F . . . slit valve according to fifth variation, 9 . . . holder, 9C . . . holder, 9a . . . interior space of holder, 11 . . . lid body, 13 . . . body, 15 . . . mouth part, 17 . . . external thread, 21 . . . ceiling wall, 23 . . . tubular outer wall part, 25 . . . tubular intermediate wall part, 27 . . . tubular inner wall part, 29 . . . discharge nozzle, 29C . . . discharge nozzle, 29a . . . nozzle flow path, 31 . . . extended part, 33 . . . internal thread, 35 . . . space (gap), 37 . . . discharge opening, 38 . . . abutting face, 39 . . . protruding part, 41 . . . concave part, 43 . . . opening part, 45 . . . diameter expanded part, 47 . . . valve sidewall part, 47a . . . leading end face of valve sidewall part, 49 . . . valve part, 49a . . . outer circumferential part of valve part, 49b . . . slit part of valve part, 49c . . . intermediate part of valve part, 51 . . . thin-walled tube part, 53 . . . outer surface of valve part, 53C . . . outer surface of valve part, 53D . . . outer surface of valve part, 54 . . . inner surface of valve part, 54C . . . inner surface of valve part, 54D . . . inner surface of valve part, 55 . . . cross-shaped slit, 55B . . . straight-line shaped slit, 57 . . . small-diameter tube part of holder, 59 . . . large-diameter tube part of holder, 60 . . . leading end face of large-diameter tube part, 61 . . . flange part of holder, 65 . . . cylindrical wall part of lid body, 67 . . . top board of lid body, 67a . . . inner surface of top board, 69 . . . latch part of lid body, 71 . . . abutting part of lid body, 75 . . . hinge, 77 . . . concave part of cap body, 79 . . . projection of lid body, 81 . . . large-diameter cylindrical wall of discharge nozzle, 83 . . . intermediate-diameter cylindrical wall of discharge nozzle, 85 . . . small-diameter cylindrical wall of discharge nozzle, 87 . . . projecting part of intermediate-diameter cylindrical wall, 89 . . . protruding part of holder, 91 . . . concave part of holder, CLq . . . content liquid, OP15 . . . opening area of mouth part,

SA29a . . . sectional area of nozzle flow path, TH47 . . . thickness of valve sidewall part, $\Phi 49$. . . outside diameter of valve part, TC49 . . . intermediate position of outer circumferential part in thickness direction, TH49a . . . largest thickness of outer circumferential part, TH49b . . . largest thickness of slit part, TH51 . . . thickness of thin-walled tube part, L51 . . . length of thin-walled tube part, L55 . . . length of slit, O . . . center point of valve part, L . . . central axis line of valve sidewall part, PR+ . . . force in discharge direction, PR- . . . force in returning direction, AR . . . air

1. A discharge cap that is attached to a mouth part of a container having an elasticity and that discharges a content liquid associated with a pressure increase in an interior space of the container due to an external force applied to the container, the discharge cap comprising:

a nozzle that forms a nozzle flow path with a sectional area smaller than an opening area of the mouth part; and

a valve that is provided on the nozzle and that is made of an elastic material softer than the nozzle, wherein

the valve includes a valve part through which the content liquid in the nozzle flow path is discharged, a tubular held part that is held on the nozzle by a holder, and a tubular connecting part that connects the valve part and the held part to each other,

the valve part includes a slit part provided with a slit opening and closing according to a difference between an atmospheric pressure and a pressure in the interior space of the container, and an outer circumferential part coupled to the connecting part, and

the slit part more easily deforms with a pressure change in the interior space than the outer circumferential part, deforms in a discharge direction of the content liquid at

a time of discharge of the content liquid, and deforms toward the nozzle flow path after the content liquid is discharged.

2. The discharge cap according to claim 1, wherein the slit part is integrally molded of a same elastic material as that of the outer circumferential part, and is thinner than the outer circumferential part.

3. The discharge cap according to claim 2, wherein the valve part includes an intermediate part between the slit part and the outer circumferential part, and the intermediate part is gradually thinned from the outer circumferential part to the slit part.

4. The discharge cap according to claim 3, wherein the valve part has an outer surface being a spherical concave face depressed toward the nozzle flow path, and an inner surface being a flat face.

5. The discharge cap according to claim 1, wherein the slit part has a thickness equal to or less than a half of a thickness of the outer circumferential part, and the slit part is located on a side nearer the nozzle flow path in relation to an intermediate position of the outer circumferential part in a thickness direction.

6. The discharge cap according to claim 1, wherein the connecting part more easily deforms with a pressure change in the interior space than the outer circumferential part, extends in the discharge direction at a time of discharge of the content liquid, and contracts in a returning direction after the content liquid is discharged.

7. The discharge cap according to claim 1, wherein the connecting part more easily deforms with a pressure change in the interior space than the outer circumferential part, and is inclined in a circumferentially inward direction of the connecting part after the content liquid is discharged.

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